



Revisiting HCN formation in Earth's early atmosphere

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ABSTRACT

Using a new photochemical model, the HCN chemistry in Earth's early atmosphere is revisited. We find that HCN production in a CH₄-rich early atmosphere could have been efficient, similar to the results of a previous study (Zahnle, 1986). For an assumed CH₄ mixing ratio of 1000 ppmv, HCN surface deposition increases from $2 \times 10^9 \text{ cm}^{-2} \text{ s}^{-1}$ at $f\text{CO}_2 = 3\%$ to more than $1 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ (30 Tg/yr) at $f\text{CO}_2 = 0.3\%$ and 1%. These conditions may well have applied throughout much of the Archean eon, 3.8–2.5 Ga. Prior to the origin of life and the advent of methanogens, HCN production rates would likely have been at $1 \times 10^7 \text{ cm}^{-2} \text{ s}^{-1}$ or lower, thereby providing a modest source of HCN for prebiotic synthesis.

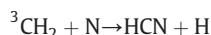
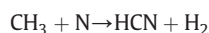
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1. Introduction

Zahnle (1986) proposed that atmosphere chemistry between methane photolysis products (CH₃ and ³CH₂) and atomic nitrogen can produce hydrogen cyanide, HCN, which is an important molecule in prebiotic chemistry and a potentially important source of fixed nitrogen to the early biosphere. This mechanism depends on the availability of methane in Earth's early atmosphere, which could have been scarce prior to the origin of life (Kasting, 1993). The methane source on prebiotic Earth is discussed in the following section. Once formed, HCN could have been transported to the surface where its hydrolysis product, ammonium ion, may have become a source of ammonia (a strong greenhouse gas) to the atmosphere (Bada and Miller, 1968). Recently, Wolf and Toon (2010) proposed that ammonia in Earth's early atmosphere could have been shielded from photolysis by fractal organic haze particles, with optical properties similar to those discovered in Titan's atmosphere (Wolf and Toon, 2010). This reopens the possibility that ammonia could have helped warm up early Earth. However, the source of ammonia was not specified in their paper. HCN chemistry in Earth's early atmosphere could conceivably have provided this missing ammonia source, helping to solve the faint young Sun problem (Sagan and Mullen, 1972). Our understanding of photochemistry has improved

over the past 25 yr, and thus it is necessary to check the validity of the Zahnle, 1986 results.

Early Earth was exposed to an intense solar EUV photon flux (Ribas et al., 2005) which would have produced atomic nitrogen through chemistry in the thermosphere and ionosphere. These N atoms would have flowed down into the lower atmosphere where they would have reacted with the byproducts of methane photochemistry (Zahnle, 1986). Kharecha et al. (2005) showed that methane concentration in early Earth atmosphere could have accumulated to ~1000 ppmv level after methanogens emerged, so methane photochemistry would have been vigorous. In such an environment HCN could then have been formed through the following two reactions:



and the HCN production rate could have reached $10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ (Zahnle, 1986), equivalent to more than 30 Tg/yr. This is comparable to or greater than the NO flux produced in volcanic lightning (Navarro-Gonzalez et al., 1998) impact plumes (Kasting, 1990), thunderstorm lightning (Navarro-Gonzalez et al., 2001), coronal discharge (Nna-Mvondo, 2005), or to reduced nitrogen produced in hydrothermal system (Brandes et al., 1998) or delivered by comets (Chyba and Sagan, 1992). Factors restricting efficient HCN formation are that: 1) the second reaction, which was predicted to be more important than the first reaction (Zahnle, 1986) could be slow

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(Navarro-Gangzalez et al., 2001); and 2) the stratosphere and mesosphere of early Earth would have to have undergone much more efficient vertical mixing than does Earth's present atmosphere in order for significant amounts of HCN to reach the surface (Zahnle, 1986). Thus, both the HCN formation rate and its surface deposition flux on early Earth have been considered as highly uncertain (Navarro-Gangzalez et al., 2001).

Formation of HCN from CH_3 was considered to be slow in Zahnle (1986) because CH_3 production from CH_4 photolysis was not included in that model. More recent experiments (Wang et al., 2000) show that 29% of methane photolysis at Lyman- α leads to methyl radicals (CH_3), and 58% leads to singlet methylene ($^1\text{CH}_2$), which can be collisionally de-excited to triplet methylene ($^3\text{CH}_2$). Thus, we expect that the formation of HCN from CH_3 and $^3\text{CH}_2$ will change from that in the Zahnle model. More importantly, it is known now that the reaction between CH_3 and atomic nitrogen produces H_2CN much more efficiently than HCN and that H_2CN can form HCN through multiple channels (Yung and DeMore, 1999). Recent observations and modeling of nitrogen-containing molecules in Titan's atmosphere (Krasnopolsky, 2009; Yelle et al., 2010 and references therein) provide the opportunity to check these reaction rates and reaction pathways. Here, we apply new knowledge to reinvestigate the HCN chemistry in Earth's early atmosphere. The key questions we ask are 1) could HCN be formed efficiently in early Earth's atmosphere? and 2) what is the relationship between HCN surface deposition flux and the CO_2 and methane mixing ratios in early Earth's atmosphere?

2. Constraints on Archean atmosphere composition

Estimating the source of HCN on the early Earth requires that one make assumptions about early atmospheric composition. The three gases that would have had the most effect on HCN are CO_2 , CH_4 , and H_2 . Oxidized gases such as O_2 would also have been important, but the concentrations of these gases can be predicted by our photochemical model.

Consider CO_2 first. The pre-industrial atmospheric CO_2 level (not including anthropogenic enhancement) was ~ 278 ppmv, or 2.78×10^{-4} bar, which we define as 1 PAL (times the Present Atmospheric Level). At 2.8 Ga, ~ 100 PAL of CO_2 would have been needed to reach the global mean surface temperature of present-day Earth (288 K), if CO_2 and H_2O were the sole greenhouse gases (Haqq-Misra et al., 2008; Kasting, 1987; von Paris et al., 2008). On the other hand, published analyses of geochemical data suggest CO_2 concentrations during the Archean in the range from ~ 3 PAL (Rosing et al., 2010) to < 100 PAL (Rye et al., 1995). The rather strict upper limit of 3 PAL atmospheric CO_2 from Rosing et al. is based on the coexistence of magnetite and siderite in banded iron-formations (BIFs). Whether or not this analysis is valid depends on whether sufficient organic matter was available to reduce the primary precipitate, ferrihydrite, to magnetite (Dauphas and Kasting, 2011; Reinhard and Planavsky, 2011). If organic matter were limiting, the pCO_2 constraint vanishes. In addition, Rosing et al. (2010) calculated above-zero surface temperature using a combination of 900 ppmv of carbon dioxide, 900 ppmv of methane, and a somewhat lower-than-present planetary albedo. However, such a high CH_4/CO_2 ratio implies that thick organic hazes should form in the stratosphere (Trainer et al., 2006) and the surface should have been cooled. Thus, it is unclear whether the carbon dioxide concentration in the Archean atmosphere could have been as low as that proposed by Rosing et al. (2010).

The 100-PAL pCO_2 constraint from Rye et al. (1995) is based on analysis of paleosols and is also subject to uncertainty (Sheldon, 2006). Indeed, Sheldon argued that if updated thermodynamic data were used, the approach in Rye et al. (1995) would yield a pCO_2 value lower than 1 PAL. These geological constraints, weak as they are, are for the late Archean and may or may not apply for the early Archean,

prior to 3 Ga. Thus, in this work we examine CO_2 concentrations between 10 and 100 PAL, recognizing that still higher pCO_2 values cannot be ruled out if the paleoarchean was as hot as some researchers have suggested (Knauth and Lowe, 2003).

Predicted atmospheric CH_4 concentrations in the distant past range from ~ 0.5 ppmv for the prebiotic Earth (Kasting, 2005) to as much as several thousand ppmv for the postbiotic Archean Earth (Kharecha et al., 2005). The estimate for prebiotic CH_4 should probably be revised upward by a factor of 10, based on new measurements of CH_4 emanating from the Lost City hydrothermal ventfield (Kelley et al., 2005; Segura et al., 2007). Even this revised estimate may be too low if serpentinization of seafloor was more common in the past, as some authors have suggested (Martin et al., 2008). After methanogens appeared, the biosphere would have been efficient at converting atmospheric hydrogen into methane, and CH_4 concentrations of hundreds to thousands of ppmv are expected.

H_2 is also important on the early Earth because, along with CH_4 , it determines the basic redox state of the atmosphere. The mixing ratio of hydrogen in Earth's early atmosphere should have been around 1000 ppmv if hydrogen escape was limited by diffusion (Kasting, 1993; Tian et al., 2005). If the hydrogen escape from early Earth was energy-limited, H_2 concentrations of the order of 10% could have been possible (Tian et al., 2005), and the methane concentration could have been significantly higher than assumed here. To demonstrate the possibility of efficient HCN formation in Earth's early atmosphere, we focus on the 1000 ppmv CH_4 cases, which are applicable to the post-biotic Earth.

3. Model description

In this work we use a stand-alone photochemical model, updated from the 1-D model of Domagal-Goldman et al. (2008), which is itself a derivative of the model of Pavlov et al. (2001). A major improvement of the present model is the inclusion of HCN and its relevant chemical species in the species list (Tables 1 and 2), and the addition of HCN-relevant chemical reactions in the reaction list (Supplemental Material). The model now includes 48 long-lived species (for which transport is considered) and 30 short-lived species (assumed to be in photochemical equilibrium), linked by 434 chemical reactions.

Most long-lived species are assigned deposition velocities at the lower boundary similar to those in Domagal-Goldman et al. (2008), based on their solubility and potential to react with surface materials. The temperature profile used in the model is shown in Fig. 1. The water vapor content in the troposphere is controlled by the temperature profile. Above the tropopause water vapor is calculated by solving the combined equations of photochemistry and transport. Both the temperature and the water vapor profiles used in the model are shown in Fig. 1. For the added HCN-relevant species, we choose high deposition velocities of the order of 1 cm/s for HNCO, NCO, and CN. Simulations show that the chemical sources and sinks of these species are in good balance and thus the deposition velocities do not affect the model results. A low deposition velocity of 7×10^{-3} cm/s is used for weakly soluble HCN (Henry's Law constant $k^0_{\text{H}} = 7.5$ mol/kg bar according to the NIST database), following Zahnle (1986). Sensitivity tests show that although the HCN mixing ratio near the surface is reduced by using higher deposition velocity, the HCN surface deposition flux remains the same, similar to the results in Zahnle (1986). The mixing ratios of CO_2 and CH_4 are fixed at the lower boundary as tunable

Table 1
List of 48 long-lived species.

O	O_2	H_2O	H	OH	HO_2	H_2O_2	H_2	CO	HCO
H_2CO	CH_4	CH_3	C_2H_6	NO	NO_2	HNO	H_2S	HS	S
SO	SO_2	H_2SO_4	HSO	S_2	NH_3	NH_2	N_2H_3	N_2H_4	$\text{CH}_3^{\ddagger}$
C_2H_5	C_2H_2	C_2H_4	C_3H_8	C_2H_3	C_3H_6	C_3H_2	CH_2CCH_2	$\text{CH}_3\text{C}_2\text{H}$	
N	Glyoxal	HCN	HNCO	NCO	CN	SO_4AER	S_8AER	HCAER	

Table 2
List of 30 short-lived species.

HNO ₂	HNO ₃	H ₂ CN	C ₂	C ₂ H	CH	NH	O ₃	O ¹ D	SO ₂
SO ₂	HSO ₃	SO ₃	S ₃	NH ₂	C ₃ H ₇	C ₃ H ₃	C	C ₃ H ₅	C ₂ H ₄ OH
CH ₃ CHO	CH ₃ CO	C ₂ H ₂ OH	CH ₂ CO	CH ₂	CH ₃ O ₂	C ₂ H ₅ CHO	CH ₃ O	HCNOH	S ₄

parameters. Most species are assigned zero effusion velocities at the upper boundary, except for H, H₂, CO, O, and N. For H and H₂, effusion velocities are calculated based on the diffusion-limited escape formula (Walker, 1977), which provides an upper limit on the escape rate, and hence a lower limit on atmospheric H₂. For CO and O, downward fluxes are provided at the upper boundary in order to parameterize the photolysis of CO₂ above the top of the model grid. Sensitivity runs show that the concentration of O near the upper boundary of the model is reduced by 10% and the O concentration below 70 km altitude remains the same when the downward fluxes of O and CO are neglected. Thus the simulation results are insensitive to the upper boundary setup of O and CO.

Zahnle (1986) showed that the thermospheric and ionospheric source of atomic nitrogen could have been between a few times $10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ and $10^{11} \text{ cm}^{-2} \text{ s}^{-1}$ during the Archean. The upper boundary of our photochemical model is below the thermosphere/ionosphere and thus the model needs the N atom downward flux as an input. The results shown in the next section are for a fixed downward N flux of $10^{11} \text{ cm}^{-2} \text{ s}^{-1}$ at the upper boundary. Sensitivity studies show that the HCN surface deposition is proportional to the N downward flux. The eddy diffusivity profile used in the model (shown in Fig. 2) is similar to that in Massie and Hunten (1981) and sensitivity runs are carried out with the tropopause minimum removed.

Around 3 Ga, it is estimated that the Sun was a factor of 3 or so brighter at Lyman- α , based on observations of solar-type nearby stars (Ribas et al., 2005). From observations of young (<0.1 Gyr old) Hyades stars, Ayres (1997) estimated that the solar radiation flux at wavelengths less than 2000 Å should be less than a factor of 2 stronger than that from the present Sun. At visible wavelengths the Sun at 3 Ga was less than 80% as luminous as today (Gough, 1981). Thus, we multiply the present-day solar spectrum by 3 at Lyman- α , by 2 at 1700 Å, by 1 at 2300 Å, by 0.75 at 4000 Å, and linearly interpolate

in between these wavelengths. The surface albedo is assumed to be zero at all UV wavelengths. Three types of aerosols (sulfate, elemental sulfur, and organic haze) are calculated using method similar to those used in previous works (Domagal-Goldman et al., 2008; Pavlov et al., 2001). Their absorption and scattering in the UV are considered in the model, but clouds are neglected.

4. Results and discussion

The model-calculated surface deposition flux of HCN as a function of CH₄ mixing ratio at 3 different CO₂ levels (0.3%, 1%, and 3%) is shown in Fig. 3. The higher the CH₄ concentration, the more HCN reaches the surface. At CH₄ concentrations lower than 100 ppmv, HCN deposition rates calculated in the model are lower than $1 \times 10^8 \text{ cm}^{-2} \text{ s}^{-1}$. At CH₄ concentrations expected for an early Earth with a thriving methanogenic biosphere ($f_{\text{CH}_4} = 1000 \text{ ppmv}$, Kharecha et al., 2005) HCN deposition could reach $1 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$, or >30 Tg/yr, which is only 3 times smaller than the biological nitrogen fixation rate on the present Earth (Gruber and Sarmiento, 1997). Sensitivity studies with different measured methane photolysis branching ratios (Table 1 in Hebrard, 2006) result in less than 10% variation of the calculated HCN deposition fluxes. Because HCN hydrolysis produces ammonia, efficient HCN formation in Earth's early atmosphere could have had a significant impact on early Earth climate, which will be an interesting future investigation.

In this work we use a surface temperature of 273 K which is at the lower limit of what is expected for the Archean. Some researchers argue that the Archean was hot (Knauth and Lowe, 2003), although this is in apparent contradiction with evidence for continental glaciation at 2.9 Ga (Kasting and Howard, 2006; Young et al., 1998, and references therein). In addition the Archean Earth also likely had less continental area and greater fraction of ocean coverage (Rosing et al., 2010). Thus it

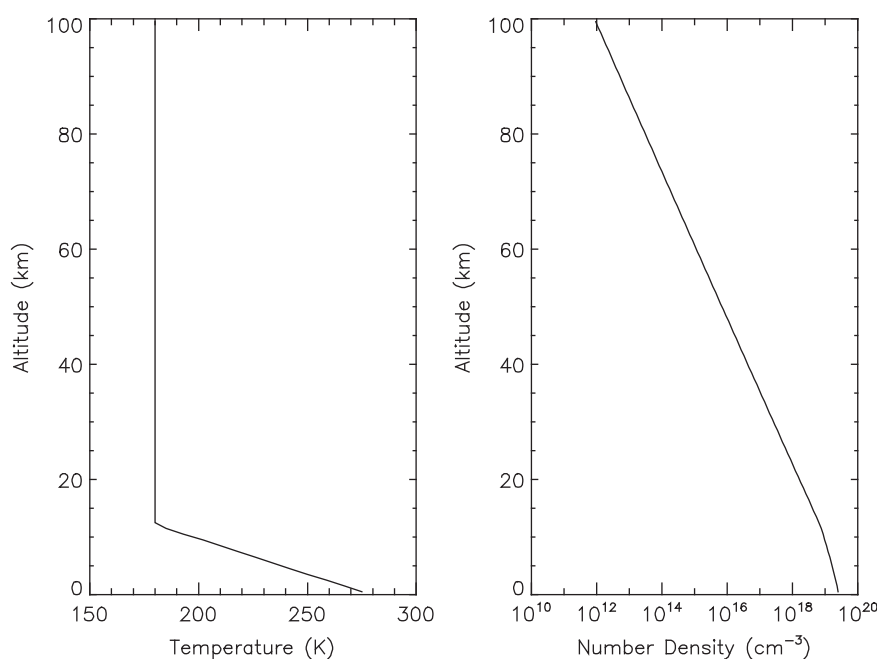


Fig. 1. Temperature and density profiles used in the photochemical model.

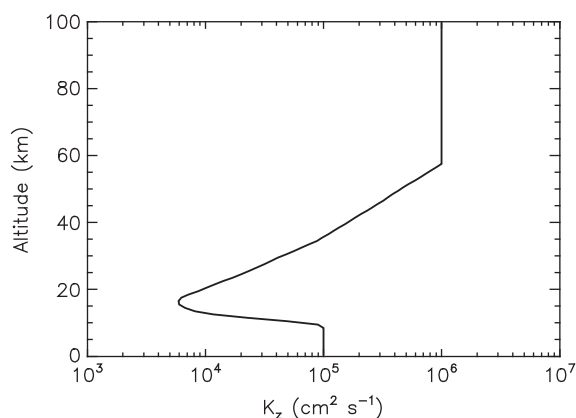


Fig. 2. Eddy diffusion coefficient profile used in the photochemical model, based on Massie and Hunten (1981).

is possible that atmospheric water vapor and rainout rate were greater than what we have assumed. To explore this effect, we increase the rainout rate in the $f\text{CO}_2 = 1\%$ and $f\text{CH}_4 = 1000$ ppmv case by a factor of 10. (This is much more than is physically possible, as currently about half of the solar radiation flux hitting the ocean surface goes into evaporating seawater. Hence, evaporation and precipitation should not increase by much more than a factor of two, regardless of how warm the surface might be.) This sensitivity study shows that enhanced rainout rate reduces HCN concentration near the surface but the combined HCN surface deposition flux (rainout plus dry deposition) remains constant, consistent with the results in Zahnle (1986). Similarly, a higher surface dry deposition velocity for HCN decreases HCN concentrations in the atmosphere but does not change the HCN deposition flux.

When compared with Zahnle's results for the present Earth eddy diffusivity profile (Fig. 11 in Zahnle, 1986), shown as the crosses for 3% CO_2 and modern eddy diffusivity cases, our new model predicts slightly less HCN deposition. Zahnle found that enhanced eddy diffusion in the middle atmosphere helped to increase HCN surface deposition. Simulations with our model using 3 times and 10 times greater eddy diffusivity in the 1% CO_2 and 0.1% CH_4 case result in 2 and 4 times more HCN surface deposition, respectively, in agreement with Zahnle's conclusion. The HCN deposition flux is increased by 10% in sensitivity runs using a constant eddy diffusivity of $10^5 \text{ cm}^2 \text{ s}^{-1}$ below 40 km. The change in the HCN flux is small because both its source and sink are at higher altitudes. On the other hand, the enhanced eddy diffusivity in the lower atmosphere does change the amount of organic haze in the atmosphere – the optical depth of organic haze drops by about 5 times. A reduction in organic haze caused by more efficient turbulent mixing could help to warm up the early Earth. We

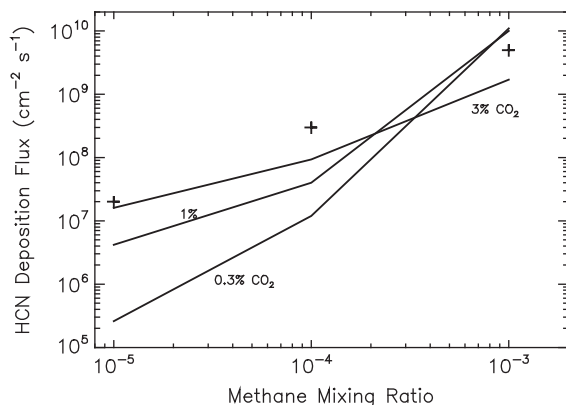


Fig. 3. Surface deposition flux of HCN as functions of methane concentration. The crosses are the HCN rainout fluxes under 3% atmospheric CO_2 in Zahnle (1986).

note that although removing atmospheric ozone may indeed increase the eddy diffusivity, it is unlikely for the stratosphere to have eddy diffusivity as high as the value observed in the current troposphere. The troposphere is convective; the stratosphere is not, even with the ozone removed. Hence, the sensitivity experiment described above probably overestimates the effect of removing ozone from the atmosphere.

Fig. 6 shows the altitude where unity optical depth is reached at different wavelengths in an atmosphere with 1% carbon dioxide and 1000 ppmv methane. Organic haze contribution to the optical depth is not included. Because Lyman- α photons do not reach the atmosphere deeper than 65 km altitude, neither methane nor HCN photolysis occurs in the lower atmosphere. Fig. 7 shows the fluxes of key species ($\text{N}(4\text{S})$, CH_4 , HCN, and CO) in the corresponding case. Both the upward flux of methane and the downward flux of HCN are constant in the lower 65 km because of the lack of short-wavelength UV photons in the lower atmosphere. The large CO downward flux ($10^{12} \text{ cm}^{-2} \text{ s}^{-1}$) near the surface is a typical behavior of early Earth atmosphere photochemistry. CO could form formate in the early oceans or be consumed by the early biosphere (Kharecha et al., 2005).

After HCN is formed in the atmosphere, it is subject to two photochemical sinks besides its downward transport to the surface: 1) UV photolysis R367 (reaction number refers to the number in the reaction table in the supplemental material): $\text{HCN} + h\nu \rightarrow \text{CN} + \text{H}$; and 2) reaction with OH radicals. Zahnle (1986) concluded that the reaction between HCN and OH should be dominated by a fast 3-body reaction to form an adduct denoted HCNOH: $\text{HCN} + \text{OH} + \text{M} \rightarrow \text{HCNOH} + \text{M}$ (R372), but he was uncertain what the actual structure of HCNOH was, nor how fast HCNOH reacts with other atmosphere species. Dean and Bozzelli (2000) point out that there are several intermediates with stoichiometry HCNOH. The likeliest HCN + OH adduct, NCHOH, establishes a bond between C and O that is difficult to break at atmospheric temperatures – we presume here that further oxidation of the adduct is inevitable (the exothermic products are $\text{HNCO} + \text{H}$ and $\text{NH}_2 + \text{CO}$), which sets a lower bound on HCN's lifetime. Our model results (not shown in Figs. 4 or 5) confirm that the OH sink is only important in the lower atmosphere and its column-integrated reaction rate is orders of magnitude smaller than that of HCN photolysis. Thus, the uncertainties involving the fate of HCNOH should not greatly impact the HCN concentration or its rate of surface deposition. Under low CH_4 conditions, higher CO_2 results in higher HCN deposition because high CO_2 leads to higher concentrations of O and O_2 , which help to conserve odd nitrogen when reacting with CN (Zahnle, 1986).

Our results differ from those of Zahnle (1986) in that the HCN deposition decreases with increasing CO_2 concentration under high CH_4 mixing ratios (1000 ppmv). To understand this result, the concentrations of important atmospheric species and chemical source and sink reactions relevant to HCN chemistry are shown in Fig. 4 for the high CH_4 + low CO_2 scenario ($f\text{CO}_2 = 0.3\%$, $f\text{CH}_4 = 0.1\%$) and in Fig. 5 for the high CH_4 + high CO_2 scenario ($f\text{CO}_2 = 3\%$, $f\text{CH}_4 = 0.1\%$). The most apparent differences in atmospheric gas concentrations are that: 1) the concentrations of O, O_2 , and CO in the low- CO_2 case are much lower than those in the high- CO_2 case (Figs. 4A and 5A); and 2) the concentrations of CH_4 photolysis derivatives, CH_3 , C_2H_2 , and C_2H_4 are more abundant in the low- CO_2 case (Figs. 4B and 5B). The reduction of O, O_2 , and CO in the low- CO_2 case is consistent with CO_2 photolysis being the major source of O and CO at the model upper boundary. Reactions of C_2H_2 and C_2H_4 with atomic oxygen (R331 and R227 in the reaction table) are the major sinks of the two radicals. Thus their abundances are higher in the low- CO_2 case.

Enhanced CH_3 concentrations lead to more efficient formation of HCN in a low- CO_2 atmosphere, since, as discussed earlier, the main channel to form HCN from CH_3 is through the reaction $\text{CH}_3 + \text{N} \rightarrow \text{H}_2\text{CN} + \text{H}$, shown as higher dashed curves in Fig. 4C and Fig. 5C, followed by H_2CN reactions with H, N, and O. Analysis of our model

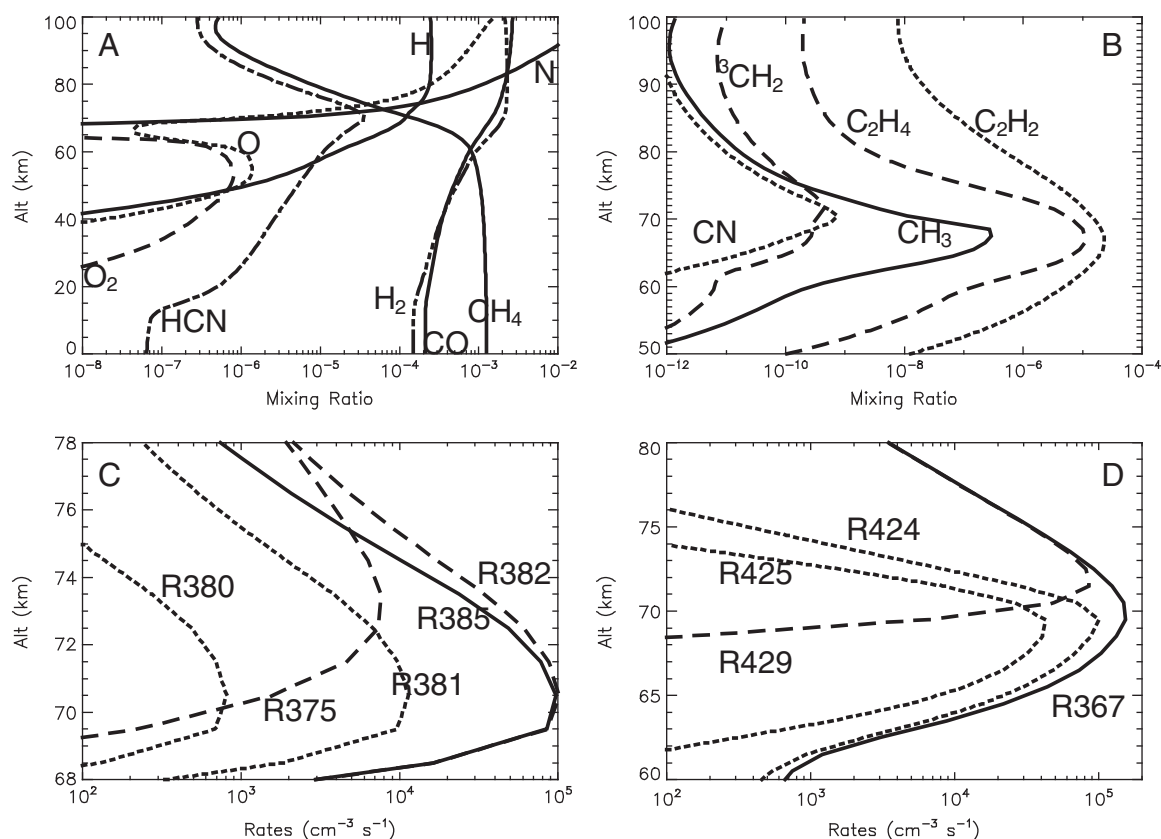


Fig. 4. Results of the photochemical model under $f\text{CO}_2 = 0.3\%$, $f\text{CH}_4 = 0.1\%$ as an example of high HCN formation scenario. The reactions in panel C and D are the following: $\text{HCN} + h\nu \rightarrow \text{CN} + \text{H}$ (R367), $^3\text{CH}_2 + \text{N} \rightarrow \text{HCN} + \text{H}$ (R375), $\text{CH}_3 + \text{N} \rightarrow \text{HCN} + \text{H} + \text{H}$ (R380), $\text{CH}_3 + \text{N} \rightarrow \text{HCN} + \text{H}_2$ (R381), $\text{CH}_3 + \text{N} \rightarrow \text{H}_2\text{CN} + \text{H}$ (R382), $\text{H}_2\text{CN} + \text{H} \rightarrow \text{HCN} + \text{H}_2$ (R385), $\text{CN} + \text{C}_2\text{H}_2 \rightarrow \text{HCN} + \text{C}_2\text{H}$ (R424), $\text{CN} + \text{C}_2\text{H}_4 \rightarrow \text{HCN} + \text{C}_2\text{H}_3$ (R425), $\text{CN} + \text{N} \rightarrow \text{N}_2 + \text{C}$ (R429).

results shows that the reaction $\text{H}_2\text{CN} + \text{H} \rightarrow \text{HCN} + \text{H}_2$ (R385) is the dominant HCN formation channel in the low- CO_2 case (solid curve Fig. 4C). R385 is also important in the high- CO_2 case, when supplemented by the reaction $\text{H}_2\text{CN} + \text{O} \rightarrow \text{HCN} + \text{OH}$ (R384) at higher altitudes (upper solid curve in Fig. 5C). The direct formation of HCN from CH_3 : $\text{CH}_3 + \text{N} \rightarrow \text{HCN} + \text{H}_2$ (R381) and $\text{CH}_3 + \text{N} \rightarrow \text{HCN} + \text{H} + \text{H}$ (R380), are less efficient (dotted curves in Figs. 4C and 5C). The reaction (R375): $^3\text{CH}_2 + \text{N} \rightarrow \text{HCN} + \text{H}$ is unimportant in the low- CO_2 case (Fig. 4C) and only marginally important in the high- CO_2 case (lower dashed curve in Fig. 5C). Thus, the uncertainties related to the reaction rate of R375 have little impact on the rate of HCN formation in the early atmosphere.

Two even more important contributions to HCN formation emerge in the high- CH_4 and low- CO_2 case. As discussed earlier, the major sink of HCN is photolysis: $\text{HCN} + h\nu \rightarrow \text{CN} + \text{H}$ (R367), shown as solid curves in Figs. 4D and 5D, followed by reaction $\text{CN} + \text{N} \rightarrow \text{N}_2 + \text{C}$ (R429, the dashed curve in Fig. 5D and the highest dashed curve in Fig. 4D) where the C–N bond is broken and odd nitrogen is destroyed. In a low- CH_4 environment, odd nitrogen can be maintained by reactions between CN and oxygen (R428 and R430, two lower dashed curves in Fig. 4D).

In a high- CH_4 environment, however, CN can be directly recycled efficiently back to HCN through the following reactions:



These reactions were not included in the Zahnle, 1986 model. Because the abundances of C_2H_2 and C_2H_4 are higher in the low- CO_2 case (Fig. 4B) than those in the high- CO_2 case (Fig. 5B), the recycling

of CN back to HCN through R424 and 425 is more efficient in the low- CO_2 case (dotted curves in Fig. 4D) than in the high- CO_2 case (dotted curves in Fig. 5D). In fact the production of HCN through the recycling of CN ($7 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$, or 200 Tg/yr, by R424 and R425) is more than the combined formation of HCN from CH_3 , H_2CN , and $^3\text{CH}_2$ ($4 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$, or 120 Tg/yr, by R385) in the high- CH_4 and low- CO_2 case. Reaction 427 ($\text{CN} + \text{CH}_4 \rightarrow \text{HCN} + \text{CH}_3$) is more than 15 and 40 times slower than R425 and R424, respectively. Thus, HCN formation in a high- CH_4 and low- CO_2 environment is helped by two effects: 1) more abundant CH_3 , which forms HCN through H_2CN efficiently; and 2) more abundant C_2H_2 and C_2H_4 , which recycles CN back to HCN efficiently.

The early oceans need to be saturated with ammonium ions in order for the HCN formed in the atmosphere to act as an efficient source of ammonia. Bada and Miller (1968) estimated that the concentration of ammonium ion in the primitive oceans should have been around 10^{-2} M . (This estimate should be regarded as speculative, because it is based on the amount of ammonia necessary to prevent deamination of aspartic acid, along with the assumption that aspartic acid was required for the origin of life. Nevertheless, it provides a number against which to compare our deposition fluxes.) All N_2 in the present terrestrial atmosphere is equivalent to 0.2 M ammonium ion in the oceans. The model calculates a downward surface flux of HCN in the order of 10^9 to $10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ after methane became abundant in early Earth atmosphere. Thus it would have taken 3 to 30 Myr to saturate the early oceans with ammonium. The subduction of oceanic crust could have been an important sink of nitrogen bearing compounds in the early Earth. The subduction flux of present Earth is estimated to be $6 \times 10^{10} \text{ moles N/yr}$ (Goldblatt et al., 2009), which is 40 times smaller than the HCN surface deposition rate we obtained. The subduction flux could have been much higher on early Earth but remains not well constrained. If additional sinks for

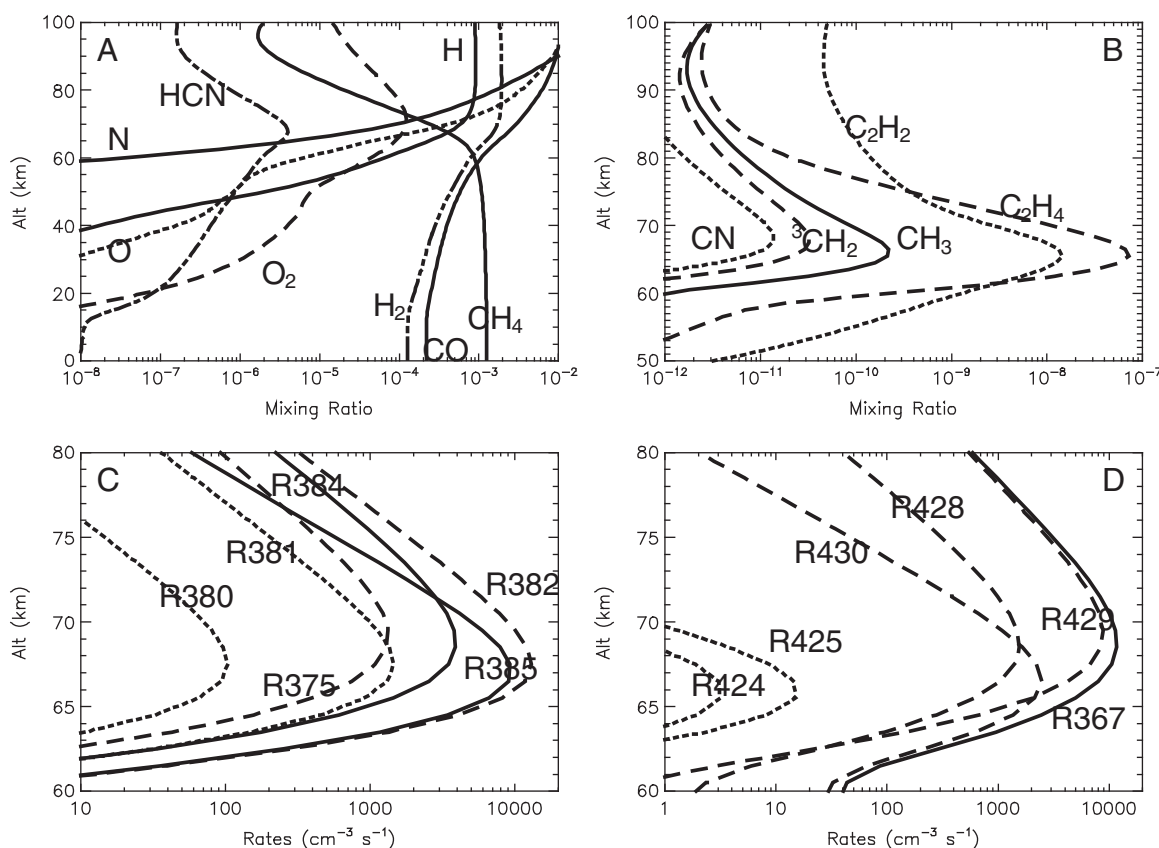


Fig. 5. Results of the photochemical model under $f\text{CO}_2 = 3\%$, $f\text{CH}_4 = 0.1\%$ as an example of low HCN formation scenario. In addition to the reactions in Fig. 4, the following reactions are added: $\text{H}_2\text{CN} + \text{O} \rightarrow \text{HCN} + \text{OH}$ (R384), $\text{CN} + \text{O} \rightarrow \text{CO} + \text{N}$ (R428), $\text{CN} + \text{O}_2 \rightarrow \text{NCO} + \text{O}$ (R430).

HCN or NH_4^+ in the early oceans are ignored, the NH_3 upward surface flux would have been the same as the HCN downward flux once the oceans were saturated with NH_4^+ .

The higher concentrations of CH_4 photolysis derivatives in high- CH_4 and relatively low- CO_2 concentration environment favor the production of hydrocarbon haze both in theoretical models (Pavlov et al., 2001) and in laboratory experiments (Trainer et al., 2006). In the case with 1000 ppmv CH_4 and 1% CO_2 , the haze extinction optical depths at 550 nm and 214 nm are less than unity and more than 5, respectively. As an alternative sink to direct surface deposition, some HCN might have become part of the haze materials. If so, what entered the Archean oceans could have included both HCN and N-bearing haze particles, some of which are insoluble while others dissolve to

make interesting prebiotic compounds. Wolf and Toon (2010) showed that the effective optical depth of haze is much smaller than the extinction optical depth. Thus efficient formation of HCN occurs at a moderate CH_4/CO_2 ratio (around 1:10), under which hydrocarbon haze remains optically thin in the visible wavelength. Because HCN deposited into paleoarchean oceans may have acted as a source of ammonia for the atmosphere, there is a window in the parameter space of methane and carbon dioxide concentrations where the ammonia source to the atmosphere could have been appreciable, perhaps allowing ammonia to contribute significantly to warming the Archean climate. The exact nature of this window requires further investigation using a coupled photochemical/climate model.

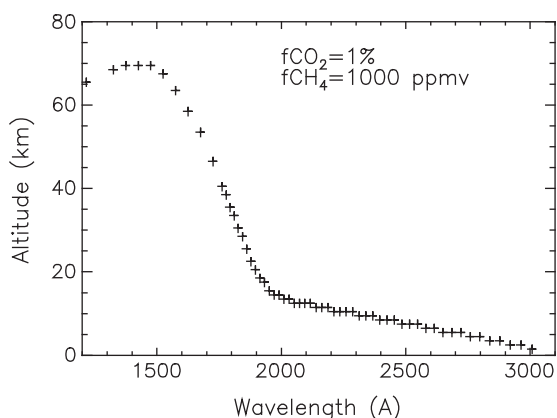


Fig. 6. The unity optical depth altitude as a function of wavelength in the $f\text{CO}_2 = 1\%$ and $f\text{CH}_4 = 0.1\%$ case. Organic haze effect on UV is not included.

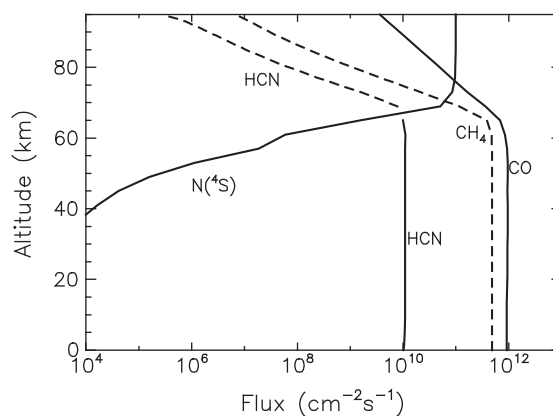


Fig. 7. Fluxes of key species in the photochemical model in the $f\text{CO}_2 = 1\%$ and $f\text{CH}_4 = 0.1\%$ case. Solid curves and dashed represent downward and upward fluxes respectively.

An HCN/NH₃ rich early Earth could have persisted as long as the methane supply to the atmosphere was strong. Solar EUV flux might have been a factor of 5 greater than that of today at 3.8 Ga and declined to 3 times that of present by 2.8 Ga based on observations of nearby solar type stars (Ribas et al., 2005). Thus the change of solar EUV flux during the Archean could have caused a long term reduction of HCN before the end of the Archean. Numerical simulations of the response of atomic nitrogen flux to the solar EUV variations will be an interesting future work. In the meanwhile, the methane source should have been stopped by the rise of oxygen around 2.4 Ga, ending the HCN supply. Methane supply to the atmosphere during the Archean may have been impacted by the environment as well. If this were the case, HCN production during the Archean could have been severely disturbed and the positive feedback between methane, ammonia, and the climate could have been important for the Archean Earth.

5. Conclusions

In summary, HCN chemistry in Earth's early atmosphere was revisited using a new photochemical model. Despite the changes in reaction pathways and reaction rates leading toward HCN formation, we confirm the results of a previous study (Zahnle, 1986) that HCN formation in a methane-rich early Earth atmosphere could have been efficient. The combination of 1000 ppmv CH₄ and 0.3% CO₂ in early Earth atmosphere could have resulted in an HCN surface deposition flux of the order of $1 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ (30 Tg/yr), which decreases to $2 \times 10^9 \text{ cm}^{-2} \text{ s}^{-1}$ when atmospheric CO₂ level increases to 3%. The implication of efficient HCN formation and deposition in early Earth atmosphere on early climate and biosphere deserve further investigation.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.epsl.2011.06.011.

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